

Mini Cases

Risk Management

2025/2026

Please upload your solutions and indicate which Mini Cases (MCs) you have solved before **June 10, 2026 at 7am**. To do so, go to the assignment section on Canvas and choose “Unit 1 Indication and Submission of MCs”. Then answer the true/false questions (Questions 1 to 6) and upload your solutions file (Question 7). **There will be no deadline extensions! Only complete submissions containing answers to the T/F questions and a solutions upload will be considered, your last complete attempt counts!** You can either write your solutions using software or scan your handwritten solutions. Please note that you may only submit one solutions file. If your solutions consists of more than one file (e.g. 1 pdf and 1 Excel file), please upload a .zip file. Uploaded files will be checked for plagiarism. The solutions in your file should be self-explanatory - not only for the instructor, but for other students as well. Indicate only Mini Cases you feel confident to present.

Important remark for the presentations: Make sure to provide a concise explanation of your solutions. In particular, when applying any formula from the lecture materials or other sources, you may be asked to explain the structure of your solutions and provide the intuition and underlying concepts behind the formula used.

Mini Case 4.1.

Consider a bond with a remaining maturity of 9.7 years. The bond is risk free and pays an annual coupon of 4%. The relevant discount rates can be found in "Mini_Case_4.1.xlsx" or "Mini_Case_4.1.Rdata".

- Calculate the present value.
- Calculate the YTM.

Hint:

Goal-Seek or a similar tool might be helpful again.

Mini Case 4.2.

Consider the bond in "Mini_Case_4.2.xlsx" or "Mini_Case_4.2.Rdata" with a remaining maturity of 8 years and a coupon of 1.5%. You already calculated the PV of 0.9978723.

- Calculate the Modified Duration and PV01 using the YTM of 0.01528373.
- What is the %-change of the present value (PV) if the interest rate level changes by -0.5% ? Calculate the change once using the MD and once by shifting the interest rates in the Excel or R file. Do you observe a difference?
- What is the %-change of the present value (PV) if the interest rate level changes by -5% ? Calculate the change once using the MD and once by shifting the interest rates in the Excel or R file. Do you observe a difference?

Mini Case 4.3.

Consider 4 zero-coupon bonds with maturities of 1, 2, 3, and 4 years, respectively, and one fixed-rate annual coupon bond with a maturity of 5 years. Assume that

the relevant yield curve is flat and equal to 2% and that the coupon bond is sold at par.

- What is the MD of a portfolio consisting of one of each bond?
- Suppose that a firm is deciding to finance its projects using one of the two portfolios:
 - (i) a portfolio consisting of the 4 zero-coupon bonds; or
 - (ii) the annual coupon bond alone.

If the MD of the assets is equal to 3.14230633, and the firm has debt equal to 2/3 of its assets. Which portfolio should the firm choose for risk management considerations?

Mini Case 4.4.

Show that:

$$\text{VAR}(P/L_{\text{Fix}}) < \text{VAR}(P/L_{\text{Float}}) \iff \text{CORR}(\text{EBIT}, I_{\text{Float}}) < \frac{\text{STDEV}(I_{\text{Float}})}{2 * \text{STDEV}(\text{EBIT})}$$

Mini Case 4.5.

Consider that you work for a firm that wants to minimize the volatility of its (pre-tax) P/L defined as EBIT minus interest payments (I) and decides between issuing a fixed- or floating-rate loan. The research department provided you with the following information: The standard deviation of EBIT is equal to 1% and the correlation between EBIT and the interest payments of the floating rate notes is equal to 0.4.

- Let $\text{STDEV}(I_{\text{Float}})$ denote the standard deviation of floating interest payments. For which levels of $\text{STDEV}(I_{\text{Float}})$ is the fixed rate optimal?

- Assume that you are not convinced that the estimated correlation is correct. However, you calculated $\text{STDEV}(I_{\text{Float}})$ and found that it is equal to 2.5%. Can you still give a recommendation on which bond minimizes the volatility of (pre-tax) P/L.

Mini Case 4.6.

Consider a firm with a modified duration of asset $\text{MD}_A = 5$ that has financed 1/2 of its assets with debt.

- Suppose the debt consists of a fixed-coupon bond maturing in 10 years bought at par and assume $r = 8\%$ for all maturities. What is the modified duration of the firm's equity?
- Suppose instead that the firm's debt consists of a floating-rate loan maturing in 5 years and paying 12M Euribor. What is the modified duration of the firm's equity?
- Suppose that interest rates increase by 5 percentage points. Calculate the percentage change in the value of the firm for both scenarios above. Which debt would you choose and why? Give an example of a (prominent) firm or bank that failed due to a combination of high duration of assets, low duration of debt and high leverage in the face of central bank interest rate increases.